Real-Time 3-D Thermal Simulation of Advanced Packages via Generative Adversarial Networks

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***Abstract*—Thermal optimization plays a crucial role in the design of advanced systems in package. Due to the large number of thermal simulations needed for full design space exploration, reductions in simulation run-time are critical. Here, we propose a data-driven approach to physics simulation by using neural networks (NNs) to cast the temperature solution process into an image-to-image translation problem. We first model the power generation map, conductivity map, and boundary conditions (BCs) into separate channels of an image. We then generate temperature solutions by training a generative adversarial network, composed of a U-Net shaped generator and a discriminator. The resultant NN model can handle diverse thermal simulation scenarios with accuracy. More importantly, our model can handle BCs, power maps, and physical package designs which are unseen during the training. Experiments show that speed wise, it enables near real-time design, providing a 2581× and 9171× speedup over a custom sparse matrix optimized finite element method and ABAQUS, respectively. Comparisons with state-of-the-art methods have demonstrated the accuracy, efficiency, and versatility of the proposed work.**

***Index Terms*—3-D packaging, convolutional NN (CNN), deep learning, heat conduction, heterogeneous integration (HI), image-to-image translation, neural network (NN), thermal management.**

I. Introduction

HETEROGENEOUS integration (HI) technologies in semiconductor packaging are key to meeting future microelectronics needs, allowing Moore's law of scaling to be sustained and even surpassed [1]. Technologies, such as Chiplets and 3-D integrated circuits (3-D ICs), provide the benefits of continued system scaling at the cost of design complexity. Along with increased power densities, the material complexity of Chiplets and 3-D ICs present new thermal, and as a result mechanical, challenges to mitigate [2].

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These factors culminate in a significantly increased need for accelerated early-stage thermal analysis and co-design in heterogeneously integrated packages [3]. The physical design of an advanced package accounting for multiphysics performance may require the thermal assessment of thousands of chiplet floorplan options using stochastic optimization algorithms [4]. Using standard techniques such as the finite element method (FEM) to solve heat conduction's partial differential equation (PDE) at this scale is not feasible. Adding to this, traditional 2-D approximations of packages are no longer adequate as 3-D ICs introduce vertically stacked heating as well as an increased sensitivity to 3-D mechanical metrics, such as thermal warpage and stress [2].

Steady-state 3-D thermal FEM simulations of the chiplet packages discussed in the proceeding sections take on the order of seconds to minutes depending on the resolution of the model and amount of software overhead [5], [6]. This brings the total simulation time for a large scale thermal design exploration of these types of packages to days or even weeks. Therefore, accelerating simulations is critical. Neural networks (NNs) have recently received much attention as a potential replacement for conventional thermal solvers due to their potential of millisecond scale evaluation times. Since Hornik et al. [7] proved that feedforward NNs could approximate any Borel measurable function, the field of NN-based methods for physics simulation has split into two distinct training methods: 1) data driven and 2) physics informed.

Data-driven networks provide a ground-truth (GT) solution during every training step and a measure of difference, typically mean-square error, is used to evaluate the network's loss. The physics-informed method developed much later. Physics-informed NNs (PINNs) optimize multilayer perceptrons (MLPs) for learning the dynamical process of a physical system. While some papers described the idea as early as the 1990s, Maziar Raissi's 2019 paper revived the idea and coined the term [8], [9]. Instead of using GT data, PINNs evaluate the governing PDE as their loss function. This idea is popular for a number of reasons, mainly that it does not require training data [10]. Additionally, arguments have been made that PINN results are more physically consistent because they directly approximate the physics instead of a simulation of said physics [11]. However, by transforming a PDE solution problem into an optimization problem, PINN-based methods are limited in accuracy and efficiency by the performance of the underlying optimizers. This dependency

TABLE I
3-D STEADY-STATE HEAT CONDUCTION NN METHODS

		Generalization Ability				
Name	***Method***	***Material***	***Power***	***Dirichlet BC***	***Neumann BC***	***Robin BC***
Wang et al. [15]	CNN	✓	✓	✗	✗	✗
Chen et al. [16]	GCN	✓	✓	✗	✓	✗
DeepOHeat [17]	PINN-Based MLP	✓[1]	✓	✗[2]	✗[2]	✗[2]
ABAQUS [5]	FEM	✓	✓	✓	✓	✓
Stack3D [6]	FEM	✓	✓	✓	✓	✓
Our Method (Static BC)	GAN	✓	✓	✗	✗	✗
Our Method (Free BC)	GAN	✓	✓	✓[3]	✓	✓

1. Possible but not included when calculating error metrics
2. Generalized value but not location
3. Constrained boundary conditions

has caused convergence difficulty when solving indefinite and ill-conditioned PDEs governing complicated physics [12]. Moreover, when the studied physics can be easily simulated, as is the case with steady-state heat conduction, the benefits of data-less training are minimized. For these reasons, a data-driven approach is adopted in this work.

Both data-driven and physics-informed methods are only useful if the produced network is able to accurately capture the entire design space. To quantify this, we define the metric of "generalization[1] ability" to be the amount of inputs to the heat conduction equation that may be freely varied by the user without the need for retraining.

Here, we present a novel framework for thermal simulation with a data-driven deep NN. We model advanced packages with power, conductivity, and BC maps as our input design features to the model. The trained network predicts 3-D thermal solutions in milliseconds, vastly accelerating the traditional 3-D thermal simulation process [5], [6]. We adopt a conditional generative adversarial network (GAN) as our learning algorithm, which interprets the whole 3-D chip simulation process as an image-to-image translation task [13]. Unlike previous data-driven works which also use NNs, our model does not require retraining to change the location and value of most BCs.

This is an extension of our previous work [14]. Here, we provide a more detailed description of the training data generation, construction of network inputs, and learning algorithm to facilitate reproduction of our network. We also elaborate on the training process and present an expanded and improved set of results. The size of our train and test datasets have been significantly increased and constraints limiting the randomness of the power generation and BC data have been eliminated. Also included is a quantitative comparison with state-of-the-art methods in terms of accuracy, efficiency, and capability.

To summarize, our contributions are as follows.

[1]Note that the term "generalization" used in this work differs from the generalization used typically in the artificial intelligence (AI)/machine learning (ML) field. Where generalization in the AI/ML field typically means adapting properly to new, previously unseen data drawn from the same distribution as the one used to create the model, generalization in our context refers to the model handling different heat conduction equation parameters, e.g, different boundary conditions (BCs), as defined freely by users, without need for retraining.

1) To the best of our knowledge, this work presents the first fully generalized network for steady-state 3-D thermal simulation of HI packages, in terms of near-arbitrary choice of BCs, power map, and conductivity.
2) The ability to nearly freely specify the location and value of Dirichlet, Neumann, and Robin conditions without the need for retraining set this work apart from all others.
3) Even with increased complexity, results show that our model exhibits an outstanding accuracy of 0.53 mean absolute percentage error (MAPEs) while maintaining a 0.0035-s evaluation time, one of the best among similar works.

II. RELATED WORKS

There have been numerous attempts to create NN-based approximators for 2-D steady-state heat conduction. 2-D simulation networks have demonstrated high levels of accuracy and robustness. Ranade et al. [18] have developed CoAE-MLSim, a novel ML approach for solving PDEs. This approach uses decomposed subdomains stitched together via flux conservation to solve the PDE. It has recently been expanded to handle heat transfer coefficients and demonstrated the ability to handle unseen power and heat transfer coefficient maps without the need for retraining [19]. Because this method employs subdomains and needs an iterative process for its evaluation, it is significantly slower than other methods. However, it is capable of handling more complex problems with minimal error. Another popular choice of method for 2-D simulation networks are convolutional NNs (CNNs). These networks, commonly employing U-nets, have the advantage of speed as they only require a single evaluation to solve the problem. This method is versatile and has been applied to simulation as well as solution enhancement [20], [21], [22]. As mentioned previously, advanced packaging, such as 3-D ICs, require 3-D thermal simulation in the design process for prediction of both thermal and mechanical behavior. However, when compared with the 2-D space, there are relatively few generalized NN models for 3-D steady-state heat conduction. A summary of the generalization ability of relevant 3-D heat conduction methods can be seen in Table I and is subsequently discussed in Section VIII.

Wang et al. [15] have presented a data-driven approach for 3-D steady-state conduction using a CNN. Their model has a single scalar field input and a single scalar field output of temperature. The model manages to have a single input field by implicitly representing all material properties and power generation with a numerical label that stays static during all training and testing. While this approach was fast due to the small size of input data, it has potential drawbacks in learning ability and generalization as it adds a layer of abstraction onto the mathematical basis for heat conduction by representing all input materials with a single number. The method's generalization ability is limited to semi-arbitrary conductivity and power generation. It is not generalized for arbitrary BCs, as BC data is not provided as a network input. Thus, retraining of the network is mandatory to handle simulations with new BCs. This method yielded an average error rate of 6.45% with an evaluation time of 0.015 s on a GeForce RTX 2080 Ti.

Also in the data-driven space, Chen et al. [16] have created a graph convolutional network (GCN) for 3-D steady-state conduction. This work seeks to decompose the thermal simulation into a compact thermal model (CTM) approximation and use the network's graph structure to encode the model. A unique and interesting contribution of this method is the use of global total power calculations over chiplets to penalize unequal energy balances during training. This method had a much slower evaluation time ranging from 0.1322 to 0.5285 s on an NVIDIA Titan RTX GPU with 24 GB of memory. The range of times corresponds to various chip simulation sizes and highlights the additional computational overhead of a GCN. This method showed excellent accuracy with a mean root-mean-square percent error (RMSPE) of just 0.8%–1.53%, but like the previous model does not allow for free choice of BCs.

In the field of PINNs, Liu et al. [17] presented an operator framework, DeepOHeat, to learn a functional map of the steady-state heat conduction PDE. The MLP network takes a power map, material properties, and positional coordinates as inputs and combines multiple branch nets with a trunk net to provide a temperature map. The approach requires homogeneous material properties in the network domain. This can be circumvented by stacking multiple domains on top of each other, but evaluation times and errors are not provided for this scenario. While only dealing with simple geometry, this method produces an impressive average of 0.02%–0.16% error with a 0.001-s evaluation time on a Tesla V100 GPU. While DeepOHeat boasts an impressive generalization ability, it falls slightly short of full generalization as the network must be retrained to change the BC locations.

III. Thermal Formulation

The strong form equations for heat conduction are shown in (1)–(3) and can be readily derived from first principles

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot \vec{q} - s \text{ in } \Omega. \tag{1}$$

$$\vec{q} = -k\nabla T \text{ in } \Omega. \tag{2}$$

$$\hat{n} \cdot (-k\nabla T - \bar{q}) = h\big(T - \bar{T}\big) \text{ on } \Gamma. \tag{3}$$

Ω denotes the domain and Γ denotes the boundary of said domain. $\vec{q}$ is the heat flux vector in W/m^2, s is domain internal heat generation in W/m^3, k is thermal conductivity with units of W/mK, and h is convection in W/m^2. Because we are only considering steady-state temperatures in this work, we assume the partial of temperature with respect to time on the left hand side of (1) to be zero. Thus, the physical properties ρ and C_p are not relevant and can be discarded. Equation (3) is written as a generalized BC that is capable of taking on the form of a Dirichlet, Neumann, or Robin condition as seen in Table II. This generalized form allows the derivation of the discretized weak form while considering convection as follows:

TABLE II
BC Forms

Boundary Condition	$\bar{\mathbf{q}}$	$\bar{\mathbf{T}}$	$\mathbf{h}$
Dirichlet	0	$\bar{T}$	$+\inf$
Neumann	$\bar{q}$	0	0
Robin	0	$\bar{T}$	h

$$\left[k\int_{\Omega}\{\nabla N\}\{\nabla N\}^T\, d\Omega + h\int_{\Gamma}\{N\}\{N\}^T\, d\Gamma\right]\{T\}$$
$$= \int_{\Omega} s\{N\}\, d\Omega - \int_{\Gamma} \bar{q}\{N\}\, d\Gamma + \int_{\Gamma} h\bar{T}\{N\}\, d\Gamma. \tag{4}$$

The dimensionality of this equation guides decisions about how we can represent the input data to our network without losing mathematical meaning. By condensing (4), one can obtain the standard linear equation

$$[K]\{T\} = \{f\}. \tag{5}$$

The solution to (5) will be used to generate training data for the network, but cannot be used as a basis for our network inputs.

A. Adaptation to Neural Networks

In order to balance both accuracy and speed of the network, it is imperative that inputs be provided in the most compact manner while maintaining mathematical relevance to the governing equations. Because of the large amount of input data needed for full 3-D simulations, scalar field inputs are chosen. This allows us to take advantage of well developed image processing AI advancements as any point in the scalar field input can be treated as a pixel of a 3-D image. The selection of scalar field inputs means that one cannot directly use the variables seen in (5) as network inputs. This is because the $[K]$ has a size of $N \times N$ when the input 3-D image has only N pixels. Thus, because of $[K]$'s size and the preprocessing time of stiffness matrix assembly, it is much more efficient to use (4) instead of (5) to construct the inputs. Immediately one may see that the internal power generation and boundary heat flux from a Neumann BC, which are the first two terms in the right-hand side of (4), can be combined into a single scalar field. However, the right-hand side convection term cannot be added. This is because there is also a convection term on the left hand side of the equation which modifies the stiffness matrix on the boundary. To combine it with the rest of the right-hand side would be equivalent to adding an extra Neumann condition as one cannot represent the off diagonals of the stiffness matrix in a scalar field. Because of this, the network inputs take the form of three scalar fields as follows.

1) Nodal power generation, of value of $\int_\Omega s\{N\}\, d\Omega - \int_\Gamma \bar{q}\{N\}\, d\Gamma$, on the domain and boundary.
2) Elemental values of conductivity, k, averaged onto nodes on the domain and boundary.
3) Elemental values of convection coefficient, h, averaged onto the boundary.

IV. Training Data Generation

To develop an accurate and useful model, it is essential to have a well-defined design space. The training data should be evenly distributed across all input space dimensions to avoid overfitting and facilitate the robust learning of the proposed network. Since the scope of thermal conductivity and power maps are confined to the design of chiplets arranged on top of an interposer covered in a mold compound, the training samples should reflect the realistic arrangements of these blocks. Typical sampling strategies for training data generation for ML models include Sobol sequence, Monte Carlo, and Latin hypercube sampling (LHS). However, an arbitrary selection of the chiplets positions will likely produce an unfeasible layout, i.e., overlapping and/or exceeding the package outline. Since direct sampling is not viable, it is necessary to devise a method to influence the sampling of variables to obtain the most uniform set of samples.

The L_p-discrepancy is a figure of merit that measures the nonuniformity of a set of n samples on a unit hypercube $C^d = [0, 1)^d$. It quantifies nonuniformity by comparing the volume of multidimensional intervals to the proportion of samples within them. Variations of L_p-discrepancy have been proposed to meet desirable characteristics, among which the centered L_2-discrepancy (CL_2) is invariant to subspace projections and certain transformations of $\mathcal{P}$, such as reflections about center planes and permutations of coordinates [23]. The first property avoids the problem where a set of samples might look evenly distributed in high-dimensional space but clustered in lower-dimensional projections. The second ensures that the discrepancy remains the same if the points are only rotated or reflected, requiring more diverse samples to decrease it further.

Let $\mathcal{L} = l_1, \ldots, l_n$ be a set of n layouts. Each layout l_i consists of m chiplets $b_{i_1}, \ldots, b_{i_m}$. To produce a dataset $\mathcal{L}$ with chiplet arrangements corresponding to nearly uniform distributed variables in the search space, we minimize

$$\begin{aligned} \min_{\mathcal{L}} \quad & CL_2(\mathcal{L}) \\ \text{s.t.} \quad & b_{i_j} \subseteq O \quad \forall j \in [1, m]\ \forall i \in [i, n] \end{aligned} \tag{6}$$

where O is the layout fixed outline, and the centered L_2-discrepancy is defined as

$$\begin{aligned} &CL_2(\mathbf{v}) \\ &= \left(\frac{13}{12}\right)^{2m} - \frac{2}{n}\sum_{i=1}^{n}\prod_{j=1}^{2m}\left(1 + \frac{\|v_{ij} - 0.5\|}{2} - \frac{\|v_{ij} - 0.5\|^2}{2}\right) \\ &+ \frac{1}{n^2}\sum_{i=1}^{n}\sum_{k=1}^{n}\prod_{j=1}^{2m}\left(1 + \frac{\|v_{ij} - 0.5\|}{2} + \frac{\|v_{kj} - 0.5\|}{2} - \frac{\|v_{ij} - v_{kj}\|}{2}\right) \end{aligned} \tag{7}$$

where v is the normalized variable of interest, in this case, the x and y center coordinates of each chiplet. To accomplish this optimization, we perform a suboptimization for the chiplets to fit in the outline. Once within, we add the layout to a pool of layouts. If the pool already has the desired n layouts, we replace the worst layout for the discrepancy metric with the new layout if it is better. Using this method, 1000 unique layouts were generated as an initial dataset, visualized in Fig. 1. After simulation of the initial 1000 layouts, all resulting inputs and outputs were rotated three times in 90° increments to produce a full dataset of 4000 samples. To prevent contamination of the testing dataset, samples were split into train, test, and validation sets prior to rotation and the rotated test and validation samples were discarded. This ensures that every layout seen in the test or validation data set was not seen during training and is itself unique from the rest of the dataset.

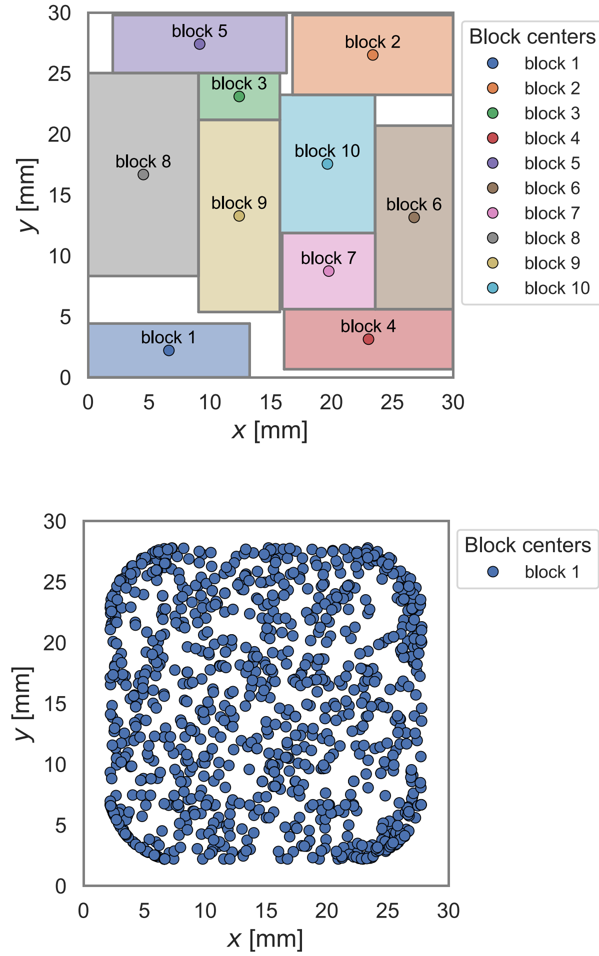

Fig. 1. Layout example with each chiplet center highlighted (above). A set of 1000 layouts optimized to minimize the centered L_2 discrepancy as in (6) displayed through chiplet 1 centers (below).

Power generation is treated as volumetric and restricted to the chiplets. Each time a new simulation is run for training data generation, all ten chiplets have their powermaps randomized. Within each chiplet, values are randomly assigned in a five by five grid. The magnitudes are evenly distributed across [0, 0.4] W/mm^3. The dimensions of the 25 boxes in the powermap's grid are an even division of the chiplet's length or width by the number of grid boxes in that dimension. In order to verify that

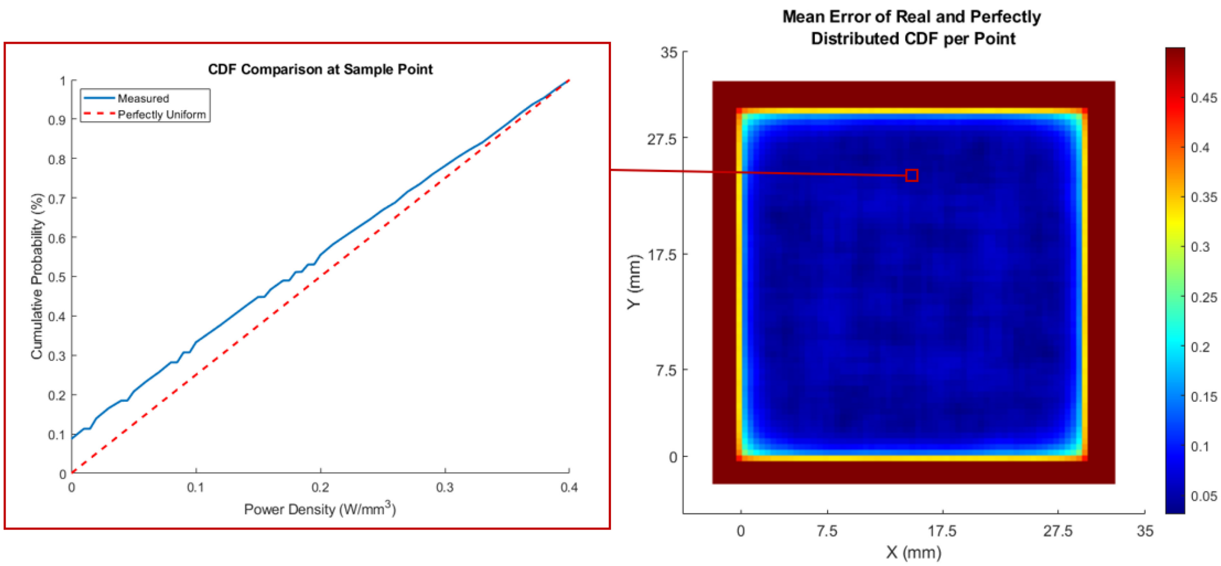

Fig. 2. Quantitative generalization metric of internal power generation. Within each pixel, 4000 power generation values are read in from the 4000 pieces of generated data and fitted to a cumulative probability distribution. It is then compared to a perfect uniform random distribution and the mean error is calculated. The theoretical value of a perfectly nonrandom and random distribution is 0.5 and 0, respectively.

this method of powermap randomization was producing well distributed training data, a comparison to an ideally distributed dataset was made. First a z-slice was taken through the chiplets in all 4000 samples. As the input image resolution is constant, 4000 power generation values were then collected at each x–y pixel location. Then, the mean error was calculated at each pixel with respect to the cumulative distribution function of a perfect uniform distribution as seen in Fig. 2. The results of this show a good distribution of power generation data in our model.

While conductivity and power generation inputs are generated to maximize randomness, the BCs have factors that limit how they can be assigned. When the mold compound encases the whole structure, BCs may be assigned to any of the six external surfaces. We support known temperature, known flux, and convection conditions. The first restriction on BC generation is fundamental to the PDE's mathematics and states that at least one known temperature or convection condition must be provided for the system to be solvable. This is because a known flux condition only defines the first derivative of temperature, leaving the value floating. Second, we must take into account the aspect ratio of the model. Because the package's geometric model is very wide in two directions and thin in the third direction, BCs defined on the top and bottom have a much larger impact than the sides. Because of this, if the required known temperature or convection condition is defined only on the sides, it will not provide an adequate path for the heat to exit the package. The result of this is an evaluated temperature in the thousands of Kelvin. This kind of result, while mathematically correct, is not physically realistic and will drastically hinder the training process because of its strong deviation from the typical temperature range. To mitigate this issue, a requirement that at least one convection or known temperature condition must be defined on either the top or bottom boundary of the package is enforced. Finally, known temperature BCs are not placed on the sides of the chips. This is because known temperature conditions defined on the top or bottom of the package as well as a side surface will form an invalid BC on the connecting edge. Additionally, heat sinks large enough to be approximated as a known temperature BC will realistically only be applied to the top of the package or bottom of the board. Thus, we allow known temperature conditions on these surfaces and not the sides to avoid any conflicts during random training data generation.

All training data is generated in the thermal mechanical advanced semiconductor analysis platform Stack3D, developed by authors Michael Smith and Dr. Ganesh Subbarayan as well as HiDaC Lab members Chung-Shuo Lee and Dr. Chun-Pei Chen [6]. Stack3D employs a symbolic representation of the package that allows automatic meshing and parsing of chiplet layouts. Our sparse matrix finite element solver constructs the system and solves for temperature in an average of 9.035 s. Using this, 1000 individual layouts with random powermaps and BCs were obtained to use for training and testing.

V. Learning Algorithm

Inspired by conditional image generation using deep NNs, we take an image-based approach to 3-D temperature solution prediction [24]. We interpret the whole process of steady-state thermal simulation of HI packages as an image-to-image translation problem. The image-to-image translation task aims to translate images from a source domain to a target domain by learning a mapping between the input and output images. The temperature map represents a target domain, while chip layouts with arbitrary physical design, power maps, and BCs represent a source domain. The mapping between two domains is learned using a training set composed of pairs of aligned cross-domain images. For successful chip design to thermal simulation translation, we applied image conditional GANs [13]. Many previous works, [24], [25], and [26], have

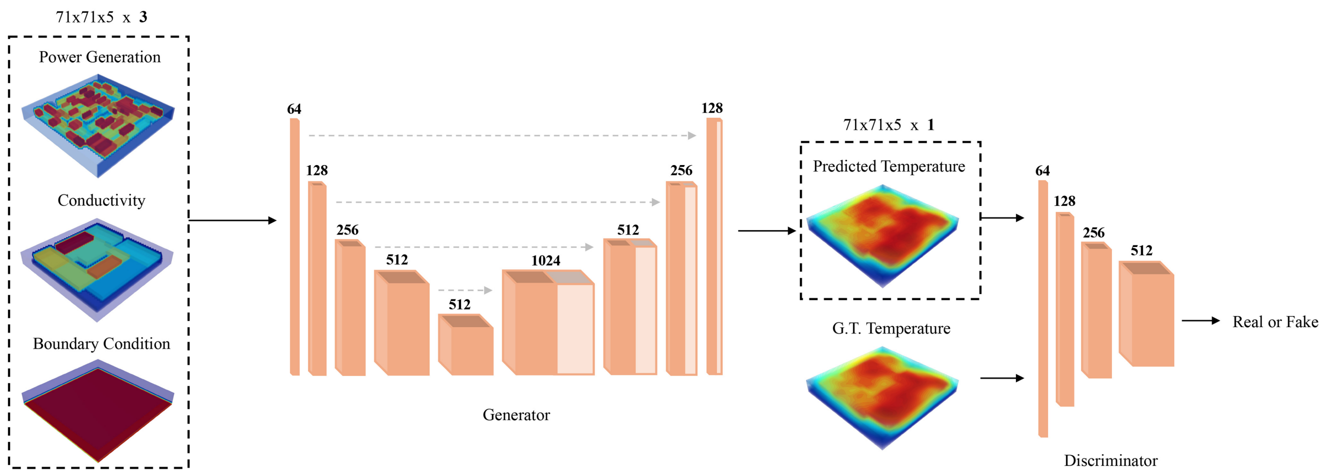

Fig. 3. Overview of the proposed image-based learning approach. 3-D shaped input conditions of power generation, conductivity, and BC, of resolution 71×71×5 (left), are stacked as respective channels of an input image, resulting in a shape of 71×71×5×3. Here, 71 corresponds to the x and y resolutions of the floor plan and 5 corresponds to the vertical resolution. These are passed to the U-Net shaped generator, which predicts a temperature solution. The discriminator distinguishes the real temperature solution from the data created by the generator. Each block in the generator and the discriminator represents different feature maps, where the number on top denotes channel dimension.

used conditional GANs for image-to-image tasks and showed promising results.

An illustration of our overall method is presented in Fig. 3. To integrate different components of the thermal model into a single 3-D image, we model the power generation map, conductivity map, and BCs as separate channels of an image. Visualization of input data is presented in the leftmost dashed box of Fig. 3. Then, the stacked image is processed with the NN to generate an output temperature solution image as presented in the rightmost dashed box in Fig. 3. For successful training of pairwise image generation, the size of output images, i.e., temperature solution, should always be kept the same as input data. In this work, we used a resolution of 71×71×5 for the input data, e.g., power map, conductivity, and BCs. The choice of 71×71×5 for the x–y–z input size means we are representing our model with a grid of 71 data points in the x direction, 71 in the y direction, and 5 in the z direction. Note that this has no relation to the physical size of the device, and is merely a resolution. The physical size of the device is set to be 35×35×0.9 mm. A single element of size 0.5×0.5 mm is adopted to accommodate 70 elements in x and y dimensions, resulting in input data of size 71×71. This input resolution was chosen because it was the smallest input data size that allowed us to reasonably represent all the features in our model. The resolution of input data to the NN can be flexible according to the problem.

When selecting a resolution for a new model one should be chosen so that the physical space that each pixel represents is at most half the size of the smallest feature in any of the model inputs. For example, if a model's power maps are the smallest feature and are composed of multiple 1 mm by 1 mm blocks of homogeneous power generation, a model resolution should be selected so that the pixels represent at most 0.5 mm in x and 0.5 mm in y.

Given image-based input conditions, we generate temperature solutions by training a conditional image generative model, composed of a U-Net [27] shaped generator and a discriminator. The generator is composed of encoding and decoding stages. As illustrated in Fig. 3, the encoding stage have four downsampling and stages. Each inner layer of the encoder is composed of leaky ReLU, convolution, and an Instance Normalization [28] function. A slope of 0.2 is used for leaky ReLU. The first layer only has convolution.

The decoder is composed of four upsampling stages. For the decoder, each inner layer is composed of ReLU, transposed convolution, and Instance Normalization. The last layer does not have Instance Normalization. For better generation we added skip connections, represented as dashed arrows in Fig. 3, to the generator between mirrored layers in the encoder and decoder. Also note that for three intermediate layers in both encoder and decoder, a Dropout ratio of 0.5 is used.

For the discriminator, we used a convolutional ImageGAN classifier, following Pix2pix [24]. As for typical cGANs training, the conditional inputs, e.g., power map, conductivity, and BCs, are also passed into the discriminator, along with the temperature. More specifically, the full-sized inputs and temperature map of size 71×71×5 are concatenated and processed by discriminator, which then outputs a single scalar value to classify the given input as real or fake. Each layer of the discriminator is composed of convolution, leaky ReLU of slope 0.2, and Instance Normalization. In the final layer, only the convolution is used, followed by a Sigmoid layer.

To handle stacks of 3D-shaped inputs we adopt 3-D, instead of the traditional 2-D, convolutional filters in our model. All input conditions, i.e., power generation map, conductivity map, and BCs, are stacked together as respective channels of an input image to the generator. To accommodate various sizes of chiplets, we adopt a size of 5×5×5 for convolutional kernels in the generator. Based on our empirical observations, with these larger-than-usual receptive fields, temperature distributions of large chiplets are better captured by our model. A size of 2×2×2 is adopted for all convolutional kernels in the discriminator. At every iteration of training, the generated temperature solution is processed by the discriminator.

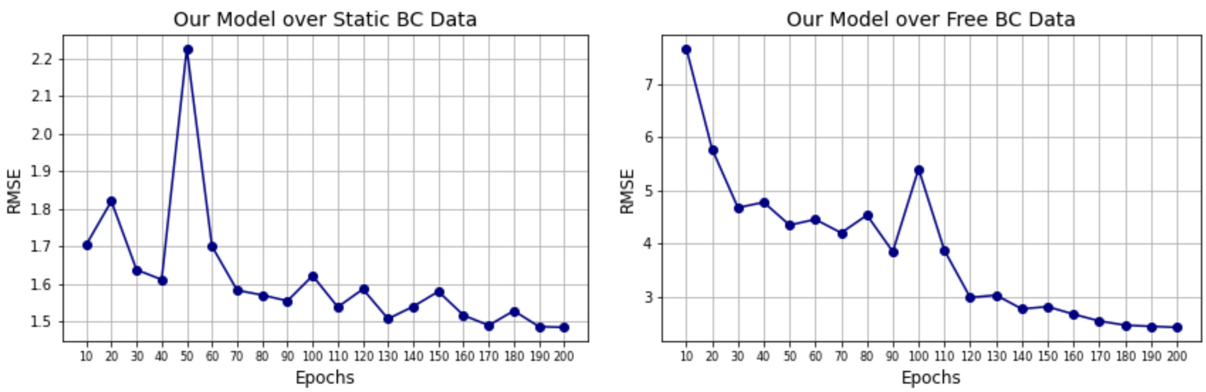

Fig. 4. Trace of RMSE scores on the validation set, with model weights saved from different training epochs. The left figure shows the RMSE score trajectory of our model trained on the dataset with uniform BCs, while the right figure shows the trajectory of our model trained on the dataset with varying BCs. We can observe that model performance on the validation set converged at around 200 epochs.

The discriminator attempts to distinguish the real temperature solution from the one generated by the model. Following the min–max optimization method for training of conditional GANs [13], the training objective of the discriminator and the generator can be reached by

$$L_{\text{cGAN}} = \arg\min_{G}\max_{D}\,[\mathbb{E}_{x,y}[\log(D(x,y))] + \mathbb{E}_{x}[\log(1 - D(G(x)))]] \tag{8}$$

where x denotes input 3-D images and y denotes the real temperature solution. Following [24], an additional L_1 loss is employed, as shown in (9), to ensure a match between the predicted and the GT temperature map

$$L_1 = \mathbb{E}_{x,y}\big[\|\|y - G(x)\|\|_1\big]. \tag{9}$$

The total training loss is defined as

$$L_{\text{total}} = L_{\text{cGAN}} + \lambda L_1 \tag{10}$$

where the choice of λ will be given in the next section based on training examples.

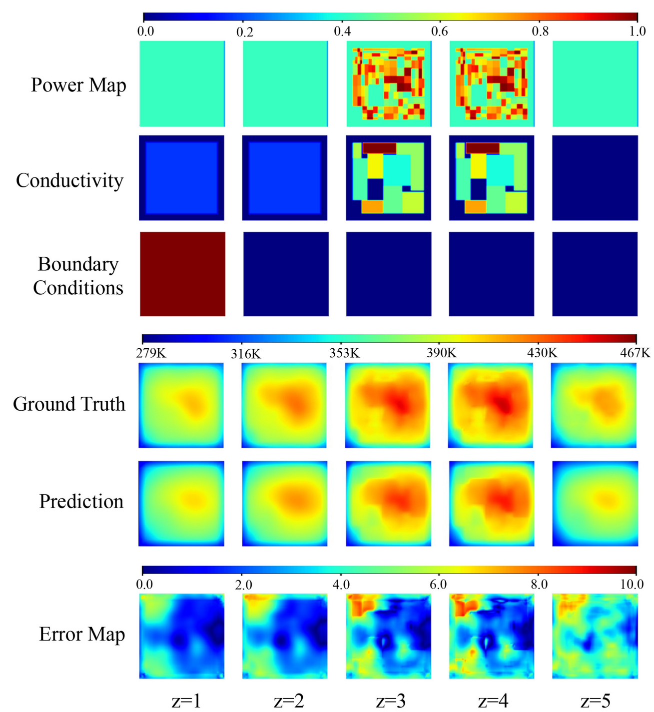

Fig. 5. 2-D visualizations of 3-D results. Vertical slices of the x–y plane from z=1 (left) to z=5 (right). The top colorbar indicates the color scale of the normalized power map [W/mm^3], conductivity [W/mm·K], and BCs [W/mm^{2}K]. The middle colorbar indicates the color scale of the GT and predicted temperature values in Kelvin. The bottom colorbar indicates the color scale of the error map, which shows absolute difference between the GT and prediction.

VI. Training

Our solver is validated under two scenarios, where the BCs are 1) static and 2) free. In the first scenario, the package's physical design and power maps are allowed to vary. In the second scenario, the previous variables as well as the Dirichlet, Neumann, and Robin BC values and locations are all allowed to vary. From these two data sets one can observe the effect that a change in dimensionality has on accuracy. The static BC dataset can also be used as a better accuracy comparison for models that do not generalize BCs. For both configurations, a total of 1000 individual layouts of thermal simulations with random power and conductivity maps are generated. Following an 8:1:1 ratio, the 1000 samples are split into sets of 800, 100, and 100 for training, validation, and testing, respectively. Additionally, all training samples were rotated three times in 90° increments to produce a final training set of 3200 samples. Note that there is no overlap between layouts from training, validation, and testing sets.

As an image-to-image translation task enforces paired datasets for training, uniform resolutions of input and output images are required. Thus, spatial resolution of all layouts are fixed to 71×71×5 in this work. For data preprocessing, all the inputs of the network, e.g., BCs, power maps, and conductivity maps, are normalized to the range of [0-1] based on the minimum and maximum values throughout the whole training data set. The temperature map is z-score normalized using global mean and global standard deviation score, so that the values have a mean of 0 and a standard deviation of 1. After prediction, the temperature maps are restored to their original scale in Kelvin.

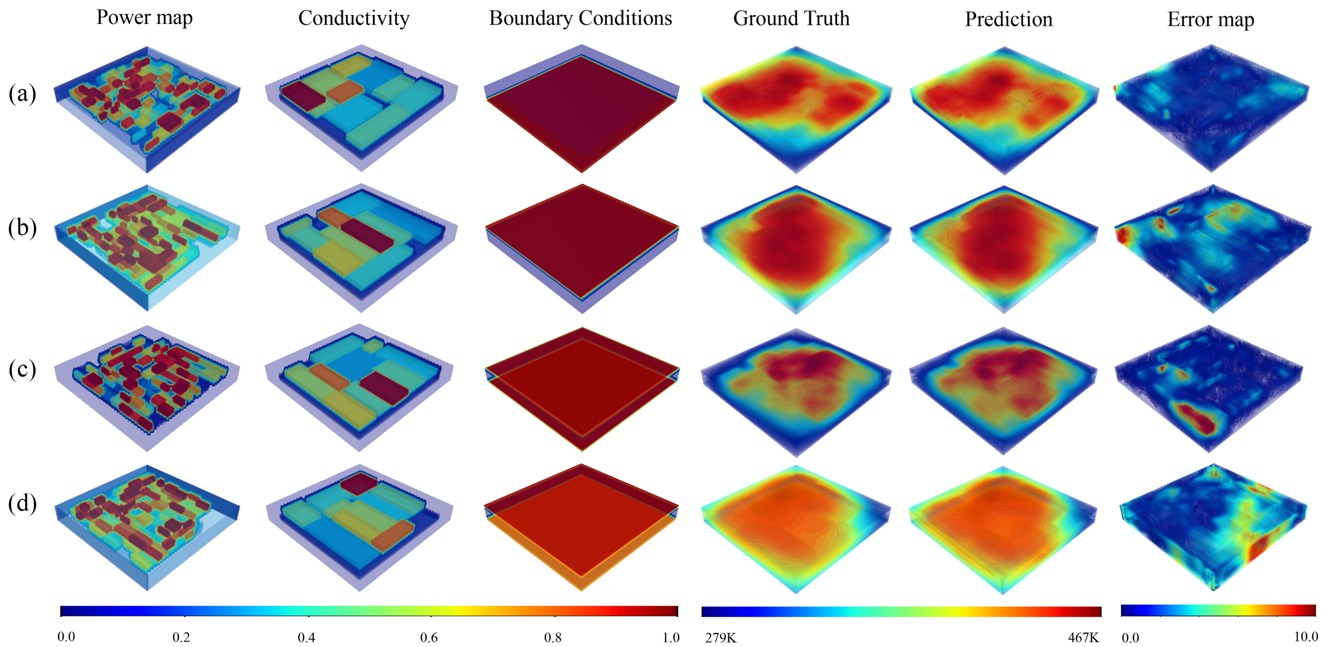

Fig. 6. Predicted thermal solutions of four chiplet packages with different layouts and input design features, (a)–(d) are presented. All packages are trained on the free BC dataset (changing BCs). The power map [W/mm^3], conductivity [W/mm $\cdot$ K], and BCs [W/mm^{2}K] are normalized to [0, 1]. The GT and predicted temperature are shown in Kelvin. The error map shows the absolute difference between the GT and prediction. All color scales are shown in the colorbars at the bottom. The BCs plots are in log scale for visualization.

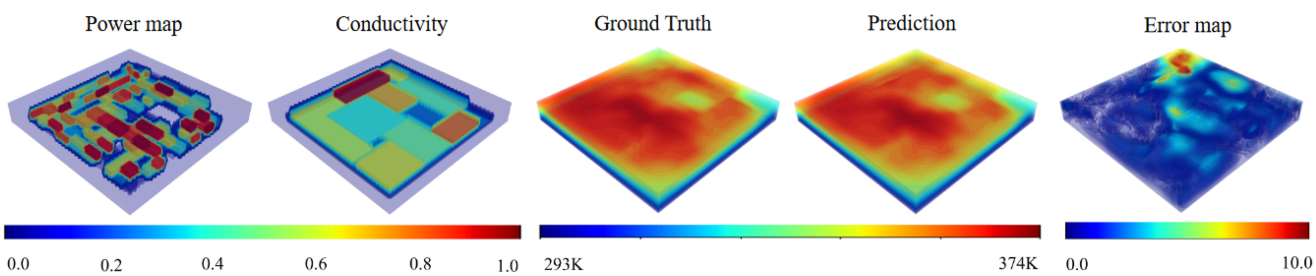

Fig. 7. Predicted thermal solutions of a chiplet package trained on static BC dataset, with uniform BCs. The power [W/mm^3] and conductivity [W/mm $\cdot$ K] maps are normalized in [0-1] scale. The GT and predicted temperature are shown in Kelvin. The error map shows absolute difference between GT and the prediction. All the color scales are shown in the colorbars at the bottom.

A. Training Parameters and Specifications

For training, λ from (10) is set to 100 as it resulted in most stable drop of training loss. After iterations of min–max optimization of the generator and discriminator, only the generator is used at the inference stage. A single RTX A5000 GPU is used for training and testing. The model is trained for 200 epochs using the Adam optimizer with momentum parameters $\beta_1 = 0.5$, $\beta_2 = 0.999$. A batch size of 1 is used throughout all training and testing, due to limited memory in the GPU card. Therefore, Instance Normalization is used instead of Batch Normalization. Instance Normalization has been shown to be effective at conditional image generation tasks by [28]. The initial learning rate is set to be 2e-4 and linearly decays for the first 100 epochs. It took 48 h of training for the full convergence of the model. The same training protocols are applied for static BC and free BC datasets. The validation loss, e.g., root-mean-squared error (RMSE) score, during training for both datasets is shown in Fig. 4. Every 20 epochs, RMSE score is calculated for the 100 validation samples and recorded. Our model converged at 200 epochs for both datasets.

VII. Results

For both static BC and free BC datasets, our model is trained on 800 training samples of unique layouts, and tested on 100 testing samples. Prior to training, the training samples are rotated into 90°, 180°, and 270° for data augmentation, resulting in total of 3200 samples We use RMSEs, peak errors (PEs), and MAPEs to measure the performance in Table III. The average inference time is also included. Note that the inference time does not include the model training time. It averages the execution time for prediction of each input sample during testing. For error calculation, the predicted temperature and GT temperature are both restored into their original scale of [279–467 K]. Our model trained on the free BC dataset is shown to have a higher error rate compared to our model trained on the static BC dataset. It is noteworthy that by

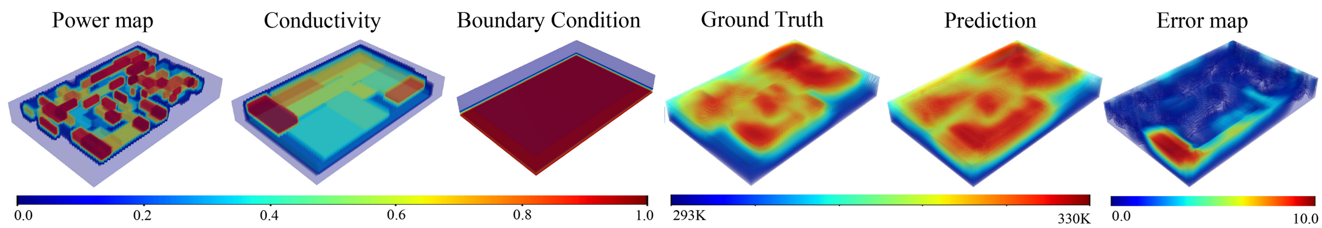

Fig. 8. Predicted thermal solutions of a chiplet with unseen dimensions and a resolution of 101×71×5, tested with the model trained on the free BC dataset. When processed by the model, the input was resized into 71×71×5 and scaled back into its original resolution after prediction. The power [W/mm^3] and conductivity [W/mm·K] maps are normalized in [0-1] scale. The GT and predicted temperature are shown in Kelvin. The error map shows absolute difference between GT and the prediction. All the color scales are shown in the colorbars at the bottom.

TABLE III
INFERENCE TIME AND AVERAGE ERRORS

Method	*RMSE [K]*	*PE [K]*	*MAPE (%)*	*Time [s]*
Ours (Free BC)	2.3049	11.1155	0.50	0.0035
Ours (Static BC)	1.6194	7.4321	0.34	0.0035

removing the BC dimension of the problem, an ~ 2× increase in accuracy is achieved.

In Figs. 5 and 6, we present 2-D and 3-D visualizations of the model inputs and outputs trained on a more challenging free BC dataset. In Fig. 6, the error maps produced by measuring absolute discrepancy of the true and predicted solutions nearly converge to zero at most locations. 3-D visualizations of the model inputs and outputs trained on the static BC dataset are shown in Fig. 7.

We also highlight the significant speed-up of our methods compared to the FEM-based solvers such as ABAQUS. A comparison is presented in Table IV. Although FEM provides impeccable accuracy, it is slow. Our method demonstrates a 9171× and 2581× speedup over ABAQUS and Stack3D, respectively, with minimum error rates.

A. Generalization to Different Input Size

Throughout all experiments, our model has been trained and tested on a package model with dimensions of 35 by 35 by 0.9 mm, represented by input data of size of 71×71×5. However, our model does not necessitate using this size or resolution when testing. With proper resizing in data preprocessing steps, input data of arbitrary shape can be resized to 71×71×5 to comply with our model. As a result, the model can accept any choice of input resolutions during testing, without need for retraining.

To empirically show that our model can predict inputs with different size from 71×71×5, we tested our network on ten samples of a new package with dimensions of 50 by 35 by 0.9 mm, represented by input data of size of 101×71×5. For generation of power map, conductivity, and BCs, the same policies used for generating the free BC datasets are applied. Input data is resized to 71×71×5 before model prediction via Bilinear interpolation. After prediction, the resultant predicted temperature map is resized into its original resolution of 101×71×5. Error rates on these ten samples are presented in Table V. For a fair comparison, we also generated a 35 by 35 by 0.9 mm version of these ten samples, represented by

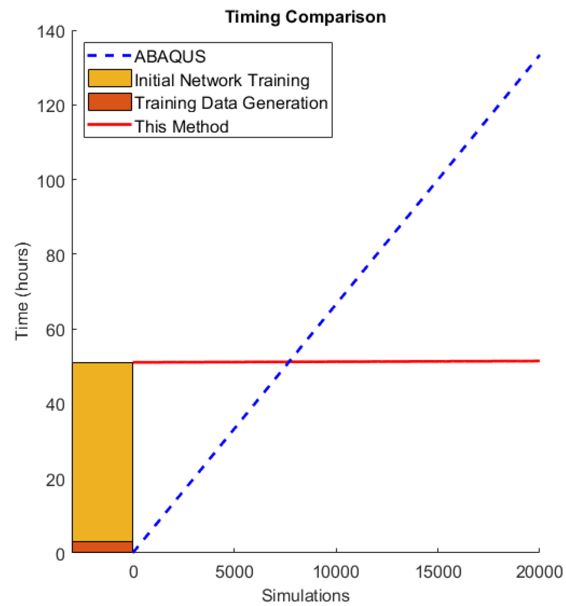

Fig. 9. Runtime comparison between a commercial software, ABAQUS, and the proposed method that takes into account the time to generate training data, train the network, and then start evaluating solutions.

input data of size of 71×71×5. Results for the 101×71×5 data are very comparable to those from the 71×71×5 inputs. A visualization is presented in Fig. 8.

VIII. BENCHMARK AGAINST EXISTING METHODS

In Table I, we evaluated the generalization ability of our solution against other methodologies [15], [16], [17], that use NN-based solutions for 3-D steady-state heat conduction. While 2-D formulations exist, only 3-D solutions were listed here because the additional dimension drastically complicates the problem. Also, from an applications standpoint, 3-D HI packages cannot be meaningfully simulated with a 2-D approximation.

It can be seen that the proposed method has the highest generalization ability of any network presented. Second to reasonable accuracy, this is the most important metric to evaluate a network's usefulness because it is the limiter to the large gains over traditional methods in evaluation time. Because training time is extremely large, typically on the order of hours to days, network solutions are functionally useless if they cannot be reused thousands of times. This point is illustrated in Fig. 9. As illustrated, only studies that require

TABLE IV
METHOD TIMING DETAILS

Method	*Training Data Generation (Hours)*	*Network Training (Hours)*	*Input Generation (Sec/Sim)*	*Simulation (Sec/Sim)*
Ours	3.03	48.02	0.0607	0.0035
ABAQUS [5]	0	0	0	24[1]
Stack3D [6]	0	0	0	9.035

1. Time calculated for simulation size used as error reference

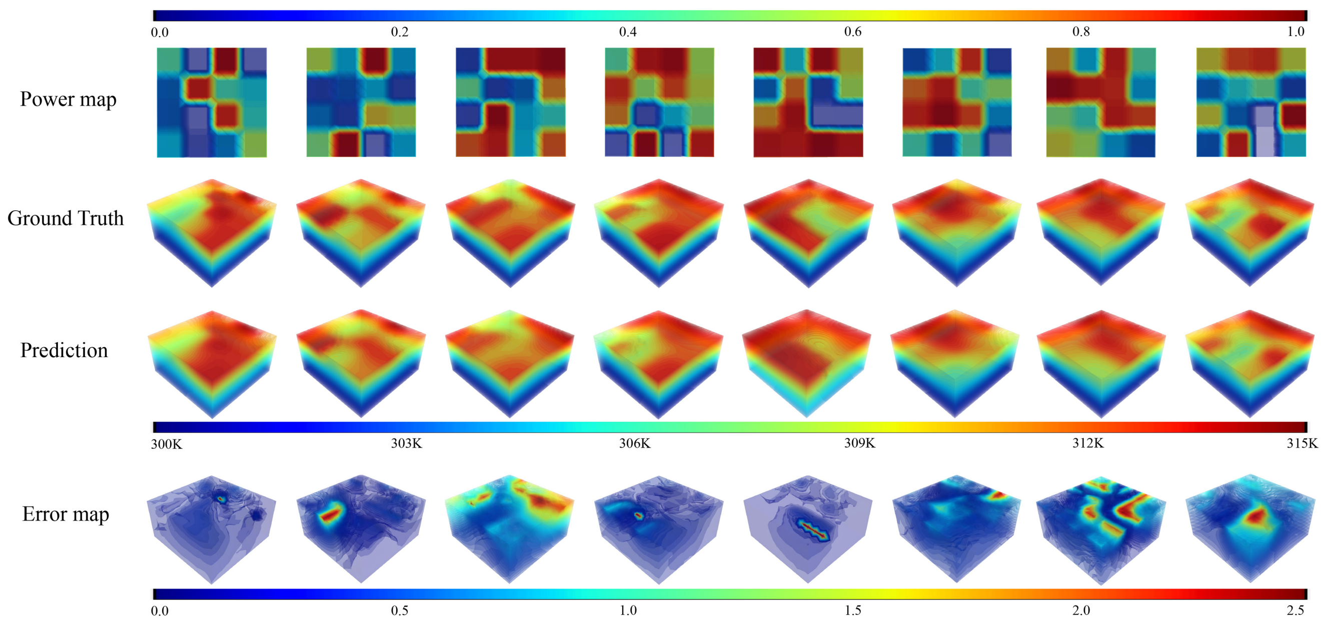

Fig. 10. Predicted thermal solutions for uniform cubes of homogeneous material with different 2-D power maps defined on the top surface. The power map [W/mm^3] is normalized to [0, 1]. The GT and predicted temperatures are shown in Kelvin. The error map shows absolute difference between the GT and prediction. Colorbars at the top, middle, and bottom show the color scales of the power map, temperature, and error map, respectively.

TABLE V
AVERAGE ERRORS ON TEN SAMPLES OF DIFFERENT INPUT SIZE

Method	*Resolution*	*RMSE [K]*	*PE [K]*	*MAPE (%)*
Ours (Free BC)	101x71x5	2.9181	14.1133	0.57
Ours (Free BC)	71x71x5	2.7552	8.0212	0.60

TABLE VI
COMPARISON WITH DEEPOHEAT [17]

Method	*Test Set Size*	*MAPE (%)*	*PAPE (%)*	*Time [s]*
Ours	800	0.1	0.45	0.001
DeepOHeat [17]	10	0.075	0.47	0.001

multiple thousands of simulations, such as our application of chiplet floor plan optimization, justify the use of an NN solution when training time is factored into the equation.

A. Comparison to DeepOHeat

We also present comparison to the DeepOHeat model, which appears to be the state of the art in terms of generalization ability and accuracy. While our model uses HI packages with ten chiplets of different conductivity values and randomized powermaps, DeepOHeat only predicts temperature for a uniform cube of homogeneous material, with a randomized powermap defined on the top surface. DeepOHeat also does not consider changing BC locations. Therefore, for fair comparison, we created a new version of our model that matches configurations of DeepOHeat. While our model has demonstrated accuracy at near full generalization, we artificially reduced the generalization to match DeepOHeat. Moreover, we attempted to follow the same data generation protocols as DeepOHeat to conform with their training and testing environments. A grid-based mesh with single cuboid geometry of resolution $21 \times 21 \times 11$ is adopted for both training and testing samples. We generated 3200 training samples and 800 testing samples, with different power maps defined on the top surface.

Comparison results with DeepOHeat are presented in Table VI. Note that our method is tested on 800 samples, which is 80$\times$ larger than the size of the test set from DeepOHeat. Nonetheless, our method showed superior peak absolute percentage error (PAPE) and comparable MAPE, which validates the accuracy of our image based learning framework. Visual results can be found in Fig. 10.

IX. CONCLUSION

For efficient learning of 3-D steady-state heat conduction, we introduced a versatile image-based learning framework. We predict 3-D thermal solutions using image-to-image translation by stacking arbitrary power generation, conductivity, and BCs into separate channels of input images. To the best of our knowledge, we are the first to propose a generalized image-based approach that can accommodate arbitrary chiplet layout, conductivity, and power generation, as well as semi-free choice of Dirichlet, Neumann, and Robin BCs. Evaluation of unseen chiplet layouts verified the accurate prediction ability of the proposed method with negligible testing time. Comparisons with existing ML methods for heat transfer have highlighted the accuracy and generality of the proposed method.

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